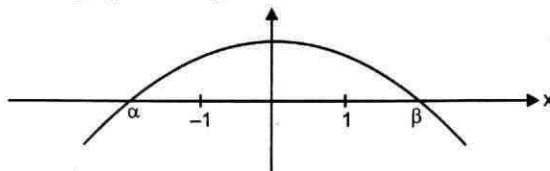


- (c) $f(x)$ takes both positive and negative values
 (d) Nothing can be said about $f(x)$
101. If the equation $x^2 + 4 + 3\cos(ax + b) = 2x$ has a solution then a possible value of $(a + b)$ equals
 (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) π
102. Let α, β be the roots of $x^2 - 4x + A = 0$ and γ, δ be the roots of $x^2 - 36x + B = 0$. If $\alpha, \beta, \gamma, \delta$ form an increasing G.P. and $A^t = B$ then the value of ' t ' equals
 (a) 4 (b) 5 (c) 6 (d) 8
103. How many roots does the following equation possess $3^{|x|} (|2 - |x||) = 1$?
 (a) 2 (b) 3 (c) 4 (d) 6
104. If $\cot \alpha$ equals the integral solution of inequality $4x^2 - 16x + 15 < 0$ and $\sin \beta$ equals to the slope of the bisector of the first quadrant, then $\sin(\alpha + \beta) \sin(\alpha - \beta)$ is equal to :
 (a) $-\frac{3}{5}$ (b) $-\frac{4}{5}$ (c) $\frac{2}{\sqrt{2}}$ (d) 3
105. Consider the functions $f_1(x) = x$ and $f_2(x) = 2 + \log_e x$, $x > 0$, where e is the base of natural logarithm. The graphs of the functions intersect :
 (a) once in $(0, 1)$ and never in $(1, \infty)$ (b) once in $(0, 1)$ and once in (e^2, ∞)
 (c) once in $(0, 1)$ and once in (e, e^2) (d) more than twice in $(0, \infty)$
106. The sum of all the real roots of equation $x^4 - 3x^3 - 2x^2 - 3x + 1 = 0$ is :
 (a) 1 (b) 2 (c) 3 (d) 4
107. If α, β ($\alpha < \beta$) are the real roots of the equation $x^2 - (k + 4)x + k^2 - 12 = 0$ such that $4 \in (\alpha, \beta)$; then the number of integral values of k equal to :
 (a) 4 (b) 5 (c) 6 (d) 7
108. Let α, β be real roots of the quadratic equation $x^2 + kx + (k^2 + 2k - 4) = 0$, then the maximum value of $(\alpha^2 + \beta^2)$ is equal to :
 (a) 9 (b) 10 (c) 11 (d) 12
109. Let $f(x) = a^x - x \ln a$, $a > 1$. Then the complete set of real values of x for which $f'(x) > 0$ is :
 (a) $(1, \infty)$ (b) $(-1, \infty)$ (c) $(0, \infty)$ (d) $(0, 1)$
110. If a, b and c are the roots of the equation $x^3 + 2x^2 + 1 = 0$, find $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$:
 (a) 8 (b) -8 (c) 0 (d) 2
111. Let α, β are the two real roots of equation $x^2 + px + q = 0$, $p, q \in R$, $q \neq 0$. If the quadratic equation $g(x) = 0$ has two roots $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}$ such that sum of roots is equal to product of roots, then the complete range of q is :

- (a) $\left[\frac{1}{3}, 3\right]$ (b) $\left(\frac{1}{3}, 3\right]$ (c) $\left[\frac{1}{3}, 3\right)$ (d) $\left(-\infty, \frac{1}{3}\right) \cup (3, \infty)$
112. If the equation $\ln(x^2 + 5x) - \ln(x + a + 3) = 0$ has exactly one solution for x , then number of integers in the range of a is :
 (a) 4 (b) 5 (c) 6 (d) 7
113. Let $f(x) = x^2 + \frac{1}{x^2} - 6x - \frac{6}{x} + 2$, then minimum value of $f(x)$ is :
 (a) -2 (b) -8 (c) -9 (d) -12
114. If $x^2 + bx + b$ is a factor of $x^3 + 2x^2 + 2x + c$ ($c \neq 0$), then $b - c$ is :
 (a) 2 (b) -1 (c) 0 (d) -2
115. If roots of $x^3 + 2x^2 + 1 = 0$ are α, β and γ , then the value of $(\alpha\beta)^3 + (\beta\gamma)^3 + (\alpha\gamma)^3$, is :
 (a) -11 (b) 3 (c) 0 (d) -2
116. How many roots does the following equation possess $3^{|x|}(|2 - |x||) = 1$?
 (a) 2 (b) 3 (c) 4 (d) 6
117. The sum of all the real roots of equation $x^4 - 3x^3 - 2x^2 - 3x + 1 = 0$ is :
 (a) 1 (b) 2 (c) 3 (d) 4
118. If α and β are the roots of the quadratic equation $4x^2 + 2x - 1 = 0$ then the value of $\sum_{r=1}^{\infty} (\alpha^r + \beta^r)$ is :
 (a) 2 (b) 3 (c) 6 (d) 0
119. The number of value(s) of x satisfying the equation $(2011)^x + (2012)^x + (2013)^x - (2014)^x = 0$ is/are :
 (a) exactly 2 (b) exactly 1 (c) more than one (d) 0
120. If α, β ($\alpha < \beta$) are the real roots of the equation $x^2 - (k + 4)x + k^2 - 12 = 0$ such that $4 \in (\alpha, \beta)$; then the number of integral values of k equals to :
 (a) 4 (b) 5 (c) 6 (d) 7
121. Let α, β be real roots of the quadratic equation $x^2 + kx + (k^2 + 2k - 4) = 0$, then the maximum value of $(\alpha^2 + \beta^2)$ is equal to :
 (a) 9 (b) 10 (c) 11 (d) 12
122. The exhaustive set of values of a for which inequation $(a - 1)x^2 - (a + 1)x + a - 1 \geq 0$ is true $\forall x \geq 2$
 (a) $(-\infty, 1)$ (b) $\left[\frac{7}{3}, \infty\right)$ (c) $\left[\frac{3}{7}, \infty\right)$ (d) None of these
123. If the equation $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ and $x^2 + (a + b)x + 36 = 0$ have a common positive root, then $b - 2a$ is equal to.
 (a) -6 (b) 22 (c) 6 (d) -22

124. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has
 (a) infinite number of real roots (b) no real root
 (c) exactly one real root (d) exactly four real roots
125. The difference between the maximum and minimum value of the function $f(x) = 3\sin^4 x - \cos^6 x$ is :
 (a) $\frac{3}{2}$ (b) $\frac{5}{2}$ (c) 3 (d) 4
126. If α, β are the roots of $x^2 - 3x + \lambda = 0$ ($\lambda \in R$) and $\alpha < 1 < \beta$, then the true set of values of λ equals :
 (a) $\lambda \in \left(2, \frac{9}{4}\right]$ (b) $\lambda \in \left(-\infty, \frac{9}{4}\right]$ (c) $\lambda \in (2, \infty)$ (d) $\lambda \in (-\infty, 2)$
127. If $2x^2 + 5x + 7 = 0$ and $ax^2 + bx + c = 0$ have at least one root common such that $a, b, c \in \{1, 2, \dots, 100\}$, then the difference between the maximum and minimum values of $a + b + c$ is :
 (a) 196 (b) 284 (c) 182 (d) 126
128. Two particles, A and B, are in motion in the xy-plane. Their co-ordinates at each instant of time t ($t \geq 0$) are given by $x_A = t$, $y_A = 2t$, $x_B = 1 - t$ and $y_B = t$. The minimum distance between particles A and B is :
 (a) $\frac{1}{5}$ (b) $\frac{1}{\sqrt{5}}$ (c) 1 (d) $\sqrt{\frac{2}{3}}$
129. If $a \neq 0$ and the equation $ax^2 + bx + c = 0$ has two roots α and β such that $\alpha < -3$ and $\beta > 2$, which of the following is always true ?
 (a) $a(a + |b| + c) > 0$ (b) $a(a + |b| + c) < 0$
 (c) $9a - 3b + c > 0$ (d) $(9a - 3b + c)(4a + 2b + c) < 0$
130. The number of negative real roots of the equation $(x^2 + 5x)^2 - 24 = 2(x^2 + 5x)$ is :
 (a) 4 (b) 3 (c) 2 (d) 1
131. The number of real values of x satisfying the equation $3|x - 2| + |1 - 5x| + 4|3x + 1| = 13$ is :
 (a) 1 (b) 4 (c) 2 (d) 3
132. If $\log_{\cos x} \sin x \geq 2$ and $0 \leq x \leq 3\pi$ then $\sin x$ lies in the interval
 (a) $\left[\frac{\sqrt{5}-1}{2}, 1\right]$ (b) $\left[0, \frac{\sqrt{5}-1}{2}\right]$ (c) $\left[\frac{1}{2}, 1\right]$ (d) none of these
133. Let $f(x) = x^2 + bx + c$, minimum value of $f(x)$ is -5 , then absolute value of the difference of the roots of $f(x)$ is :
 (a) 5 (b) $\sqrt{20}$
 (c) $\sqrt{15}$ (d) Can't be determined
134. Sum of all the solutions of the equation $|x - 3| + |x + 5| = 7x$, is :
 (a) $\frac{6}{7}$ (b) $\frac{8}{7}$ (c) $\frac{58}{63}$ (d) $\frac{8}{45}$

135. Let $f(x) = x^2 + \frac{1}{x^2} - 6x - \frac{6}{x} + 2$, then minimum value of $f(x)$ is :
 (a) -2 (b) -8 (c) -9 (d) -12
136. If $a + b + c = 1$, $a^2 + b^2 + c^2 = 9$ and $a^3 + b^3 + c^3 = 1$, then the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ is :
 (a) $\frac{2}{3}$ (b) 5 (c) 6 (d) 1
137. If roots of $x^3 + 2x^2 + 1 = 0$ are α, β and γ , then the value of $(\alpha\beta)^3 + (\beta\gamma)^3 + (\alpha\gamma)^3$, is :
 (a) -11 (b) 3 (c) 0 (d) -2
138. If $x^2 + bx + b$ is a factor of $x^3 + 2x^2 + 2x + c$ ($c \neq 0$), then $b - c$ is :
 (a) 2 (b) -1 (c) 0 (d) -2
139. The graph of quadratic polynomial $f(x) = ax^2 + bx + c$ is shown below



- (a) $\frac{c}{a}|\beta - \alpha| < -2$ (b) $f(x) > 0 \forall x > \beta$ (c) $ac > 0$ (d) $\frac{c}{a} > -1$
140. If $f(x) = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$, then complete solution of $0 < f(x) < 1$, is :
 (a) $(-\infty, \infty)$ (b) $(0, \infty)$ (c) $(-\infty, 0)$ (d) $(0, 1) \cup (2, \infty)$
141. If α, β, γ are the roots of the equation $x^3 + 2x^2 - x + 1 = 0$, then value of $\frac{(2-\alpha)(2-\beta)(2-\gamma)}{(2+\alpha)(2+\beta)(2+\gamma)}$, is:
 (a) 5 (b) -5 (c) 10 (d) $\frac{5}{3}$
142. If α and β are roots of the quadratic equation $x^2 + 4x + 3 = 0$, then the equation whose roots are $2\alpha + \beta$ and $\alpha + 2\beta$ is :
 (a) $x^2 - 12x + 35 = 0$ (b) $x^2 + 12x - 33 = 0$ (c) $x^2 - 12x - 33 = 0$ (d) $x^2 + 12x + 35 = 0$
143. If a, b, c are real distinct numbers such that $a^3 + b^3 + c^3 = 3abc$, then the quadratic equation $ax^2 + bx + c = 0$ has
 (a) Real roots (b) At least one negative root
 (c) Both roots are negative (d) Non real roots
144. If the equation $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ and $x^2 + (a+b)x + 36 = 0$ have a common positive root, then $b - 2a$ is equal to.
 (a) -6 (b) 22 (c) 6 (d) -22

145. Consider the equation $x^3 - ax^2 + bx - c = 0$, where a, b, c are rational number, $a \neq 1$. It is given that x_1, x_2 and $x_1 x_2$ are the real roots of the equation. Then $x_1 x_2 \left(\frac{a+1}{b+c} \right) =$
- (a) 1 (b) 2 (c) 3 (d) 4
146. The exhaustive set of values of a for which inequation $(a-1)x^2 - (a+1)x + a-1 \geq 0$ is true $\forall x \geq 2$.
- (a) $(-\infty, 1)$ (b) $\left[\frac{7}{3}, \infty \right)$ (c) $\left[\frac{3}{7}, \infty \right)$ (d) None of these
147. The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$
- (a) 2 (b) 4 (c) 1 (d) 3
148. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has
- (a) infinite number of real roots (b) no real root
(c) exactly one real root (d) exactly four real roots
149. If α, β are the roots of the quadratic equation $x^2 - 2(1 - \sin 2\theta)x - 2\cos^2 2\theta = 0$, ($\theta \in R$) then the minimum value of $(\alpha^2 + \beta^2)$ is equal to :
- (a) -4 (b) 8 (c) 0 (d) 2
150. If the equation $|\sin x|^2 + |\sin x| + b = 0$ has two distinct roots in $[0, \pi]$; then the number of integers in the range of b is equals to :
- (a) 0 (b) 1 (c) 2 (d) 3
151. If $a \neq 0$ and the equation $ax^2 + bx + c = 0$ has two roots α and β such that $\alpha < -3$ and $\beta > 2$. Which of the following is always true ?
- (a) $a(a + |b| + c) > 0$ (b) $a(a + |b| + c) < 0$
(c) $9a - 3b + c > 0$ (d) $(9a - 3b + c)(4a + 2b + c) < 0$
152. If α, β are the roots of the quadratic equation $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + px - r = 0$ then $(\alpha - \gamma)(\alpha - \delta)$ is equal to :
- (a) $q + r$ (b) $q - r$ (c) $-(q + r)$ (d) $-(p + q + r)$
153. Complete set of solution of $\log_{1/3}(2^{x+2} - 4^x) \geq -2$ is :
- (a) $(-\infty, 2)$ (b) $(-\infty, 2 + \sqrt{13})$ (c) $(2, \infty)$ (d) None of these